

Some Epidemiological Data on Influenza in Germany

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An explanatory summary of the databases collected for use in the Influenza Modeling Experiment is shortly presented here. They try to cover several dimensions that are directly or indirectly related to our study phenomenon, such as: Syndromic and virological surveillance, demographic data, climatic parameters, vaccination coverage and effectiveness, Individual social contact and population mobility. With respect to the nomenclature used, the WHO (WHO 2017) global influenza surveillance standards define surveillance case definitions for influenza-like illness (ILI), severe acute respiratory infections (SARI), Acute Respiratory Infection (ARI), Intensive Care Unit (ICU), Medically Attended Acute Respiratory Illness (MAARI) and Pneumonia and Influenza (P&I). Regarding the strains analyzed, the nomenclature is: AH1N1 (post-pandemic 2009), AH2N3, Bv (B-Victoria), By (B-Yamagata), B (all subtypes) and Influenza (all types) and relative to the age-groups, we have: A1 (0-4), A2 (5-14), A3 (15-34), A4 (35-59) and A5 (60+).

1.1 Epidemiological Data (Syndromic and Virological Surveillance)

Database	Source	Frequency	Period	Variables
FluID.csv	FluID	Weekly	1996-2023	n=38

FluID is a global platform for data sharing that links regional influenza epidemiological data into a single global database and provides connections between existing databases and can also be used to directly enter data by the national focal points through a web-based interface. It complements the existing virological data collection tool and accommodates both qualitative and quantitative data which facilitates the tracking of global trends, spread, intensity, and impact of influenza. In **Fig.1** we present the temporal evolution of the variables selected from the original data set, as well as a brief description of them in **Tab.1**.

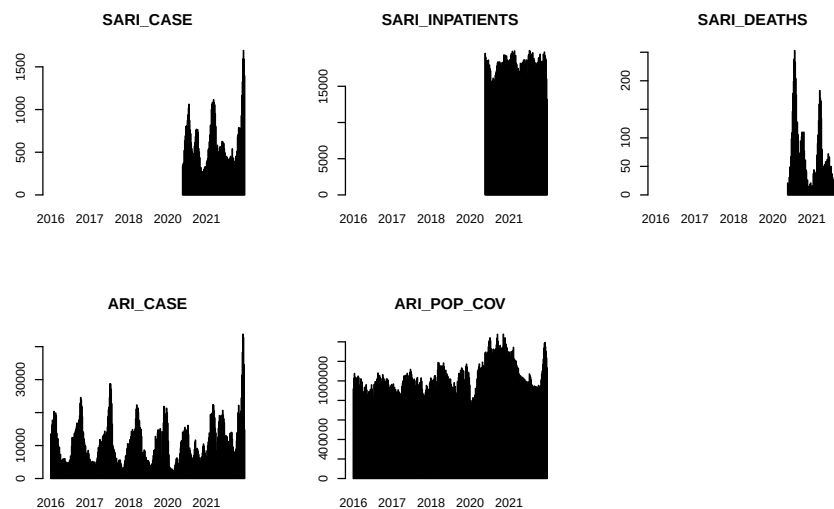


Fig.1 Reported number of weekly SARI cases, SARI hospitalized patients, SARI deaths, ARI cases, and ARI outpatient visits during the period 2016-2022.

Variable	Type	Short Description
ISO_YEAR	integer	Year (ISO 8601)
ISO_WEEK	integer	Week (ISO8601)
AGE-GROUP_CODE	string	Age Group
SARI_CASE	integer	Number of reported SARI cases
SARI_INPATIENTS	integer	Number of SARI inpatients
SARI_DEATHS	integer	Number of SARI deaths
ARI_CASE	integer	Number of reported ARI cases
ARI_POP_COV	integer	Population covered for the ARI sites

Tab.1 Short description for the variables from the FluID dataset.

Data ID	Source	Frequency	Period	Variables
FluNet.csv	FluNET	Weekly	1995-2023	n=49

Flunet is a global web-based tool for influenza virological surveillance first launched in 1997. The virological data entered into FluNet, e.g. number of influenza viruses detected by subtype, are critical for tracking the movement of viruses globally and interpreting the epidemiological data. The data at country level are publically available and updated weekly and the results are presented in various formats including tables, maps and graphs and provides the number of weekly infected people for different viral types, including Influenza, during the period 1995-2023, from Sentinel¹ and Non-Sentinel² Sources. In **Fig.2** (Sentinel Source) and **Fig.3** (Non-Sentinel Source), we present the temporal evolution of some of the variables selected from the original data set, as well as a brief description of them in **Tab.2**.

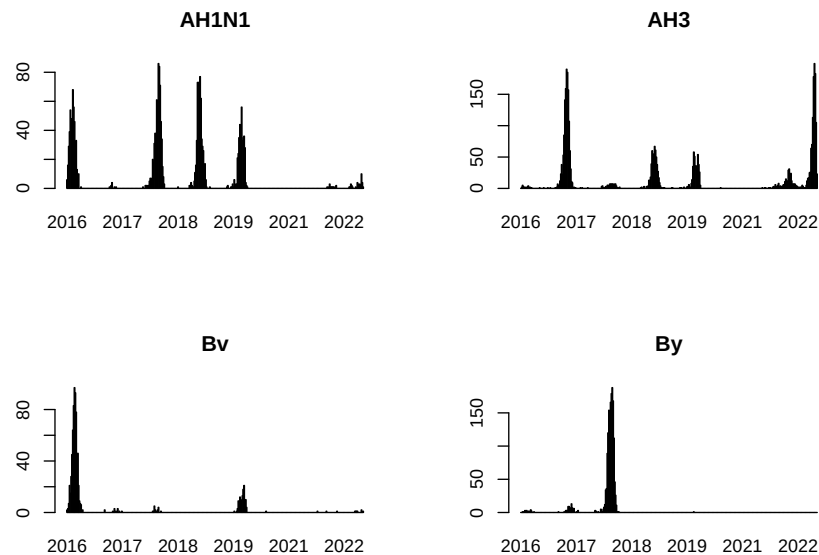


Fig.2 Number of detected cases for the influenza lineages AH1N1, AH3, Bv and By during the period 2016-2022 (Sentinel Source).

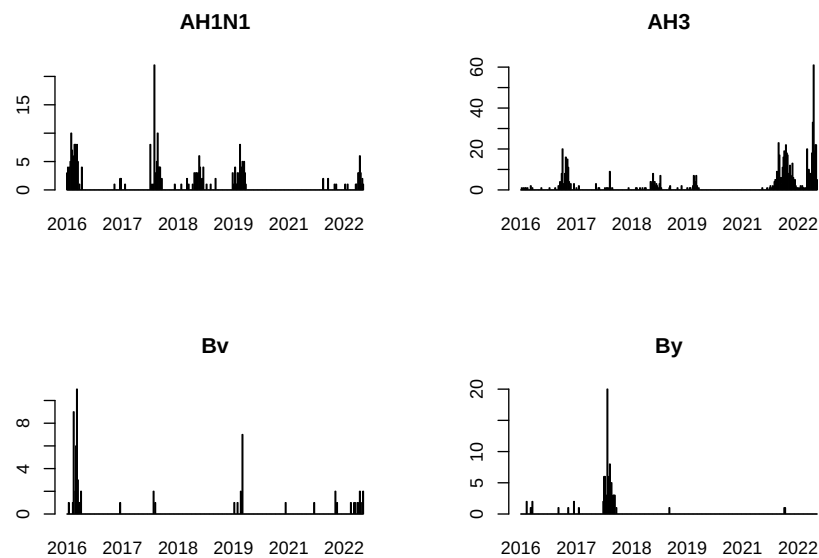


Fig.3 Number of detected cases for the Influenza lineages AH1N1, AH3, Bv and By during the period 2016-2022 (Non-Sentinel Source)

Variable	Type	Description
ISO_YEAR	integer	Year (ISO 8601)
ISO_WEEK	integer	Week (ISO8601)
S-AH1N1	integer	Sentinel Number of A(H1N1)pdm09 detections
S-AH3	integer	Sentinel Number of A(H3) detections
S-Bv	integer	Sentinel Number of B (Victoria lineage) detections
S-By	integer	Sentinel Number of B (Yamagata lineage) detections
S-INFO_A	integer	Sentinel Number of influenza A detections (all subtypes)
S-INFO_B	integer	Sentinel Number of influenza B detections (all subtypes)

1 Sentinel surveillance systems with high-quality data (FluNet information).

2 Other systems apart from sentinel surveillance (FluNet Information).

S-INFO_ALL	integer	Sentinel Number of influenza detections (all types)
NS-AH1N1	integer	Non-Sentinel Number of A(H1N1)pdm09 detections
NS-AH3	integer	Non-Sentinel Number of A(H3) detections
NS-Bv	integer	Non-Sentinel Number of B (Victoria lineage) detections
NS-By	integer	Non-Sentinel Number of B (Yamagata lineage) detections
NS-INFO_A	integer	Non-Sentinel Number of influenza A detections (all subtypes)
NS-INFO_B	integer	Non-Sentinel Number of influenza B detections (all subtypes)
NS-INFO_ALL	integer	Non-Sentinel Number of Influenza detections (all types)

Tab.2 Short description for the variables from the FluNet dataset.

Data ID	Source	Data Collection Frequency	Period	Variables
RKI-D1.csv RKI-D2.csv RKI-D3.csv	SurvStat@RKI 2.0	Weekly and Annually	2001-2023	n=4

[SurvStat@RKI 2.0](#) offers the possibility of querying a simplified database of cases of illness and evidence of pathogens that are notifiable according to the Infection Protection Act (IfSG) and are transmitted to the Robert Koch Institute (RKI). Tables and graphics can be created according to your own needs using individually created queries or by editing predefined example queries. This database provides the number of weekly Influenza cases by age group and strain, for the period 2001-2023. In **Fig.4**, **Fig.5** and **Fig.6**, we present the temporal evolution of some of the selected variables from the original data set, as well a brief description of them respectively the in **Tab.3**, **Tab.4** and **Tab.5**.

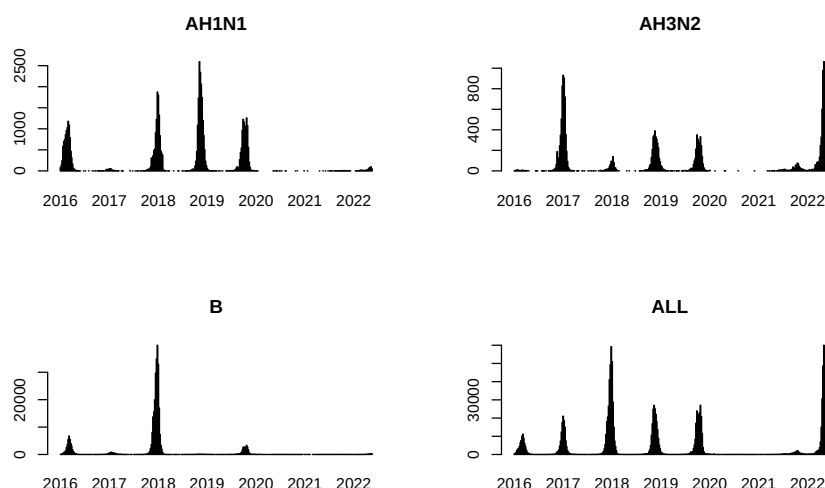
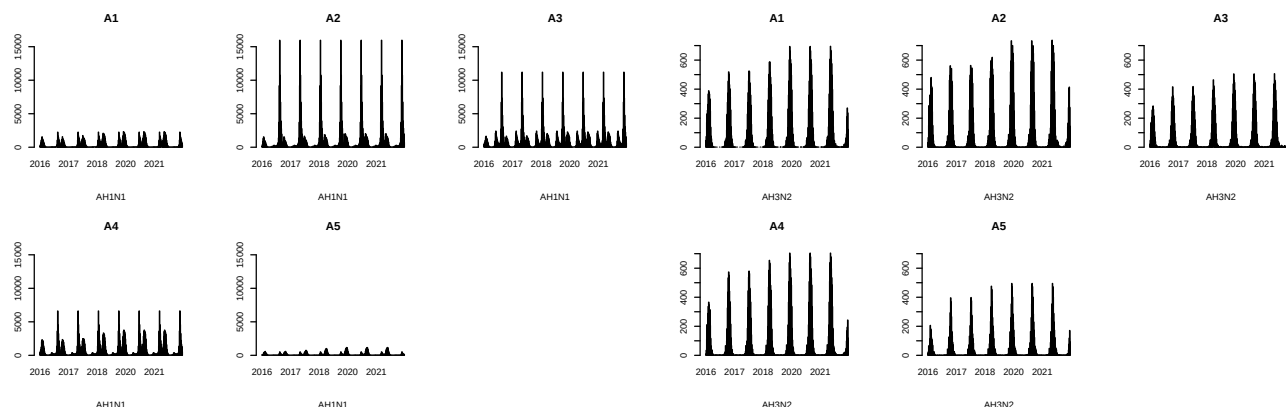


Fig.4 Number of weekly cases for AH1N1, AH3N2, B (all subtypes) and Influenza (all types) during the period 2016-2022.

Variable	Type	Description
YEAR	integer	Year (ISO 8601)
WEEK	integer	Week (ISO 8601)
AH1N1	integer	Number of AH1N1 detections
AH3N2	integer	Number of AH3N2 detections
B	integer	Number of B detections (all subtypes)
ALL	integer	Number of Influenza detections (all types)

Tab.3 Short description for the selected variables from the RKI-D1 dataset.



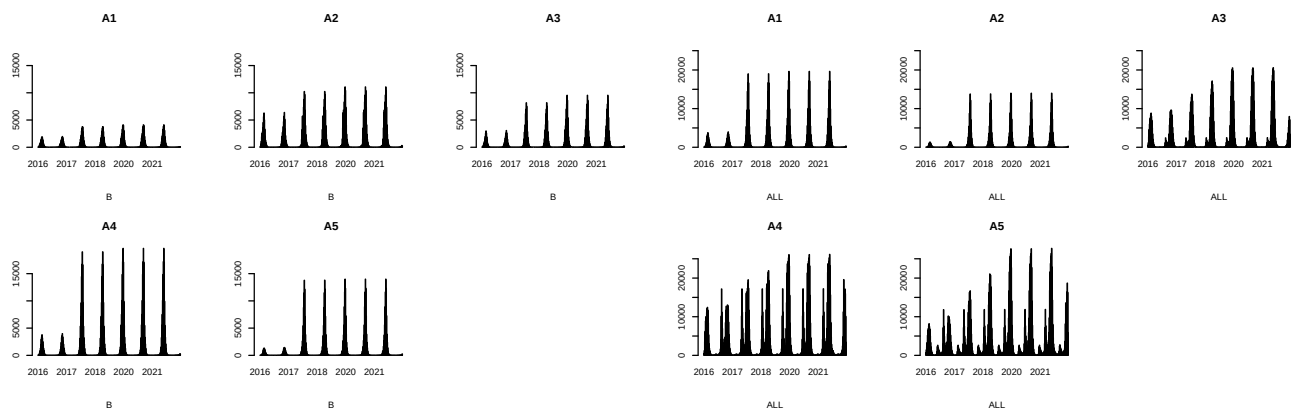


Fig.5 Number of weekly cases for AH1N1 (topleft), AH3N2 (topright), B (all subtypes, bottomleft) and Influenza (all types, bottomright) by age-groups A1-A5 for the period 2016-2022.

Variable	Type	Description
YEAR	integer	Year (ISO 8601)
WEEK	integer	Week (ISO 8601)
AH1N1	integer	Number of AH1N1 weakly cases by age-groups A1-A5
AH3N2	integer	Number of AH3N2 weakly cases by age-groups A1-A5
B	integer	Number of B (all subtypes) weakly cases by age-groups A1-A5
ALL	integer	Number of Influenza (all types) weakly cases by age-groups A1-A5

Tab.4 Short description for the selected variables from the RKI-D2 dataset.

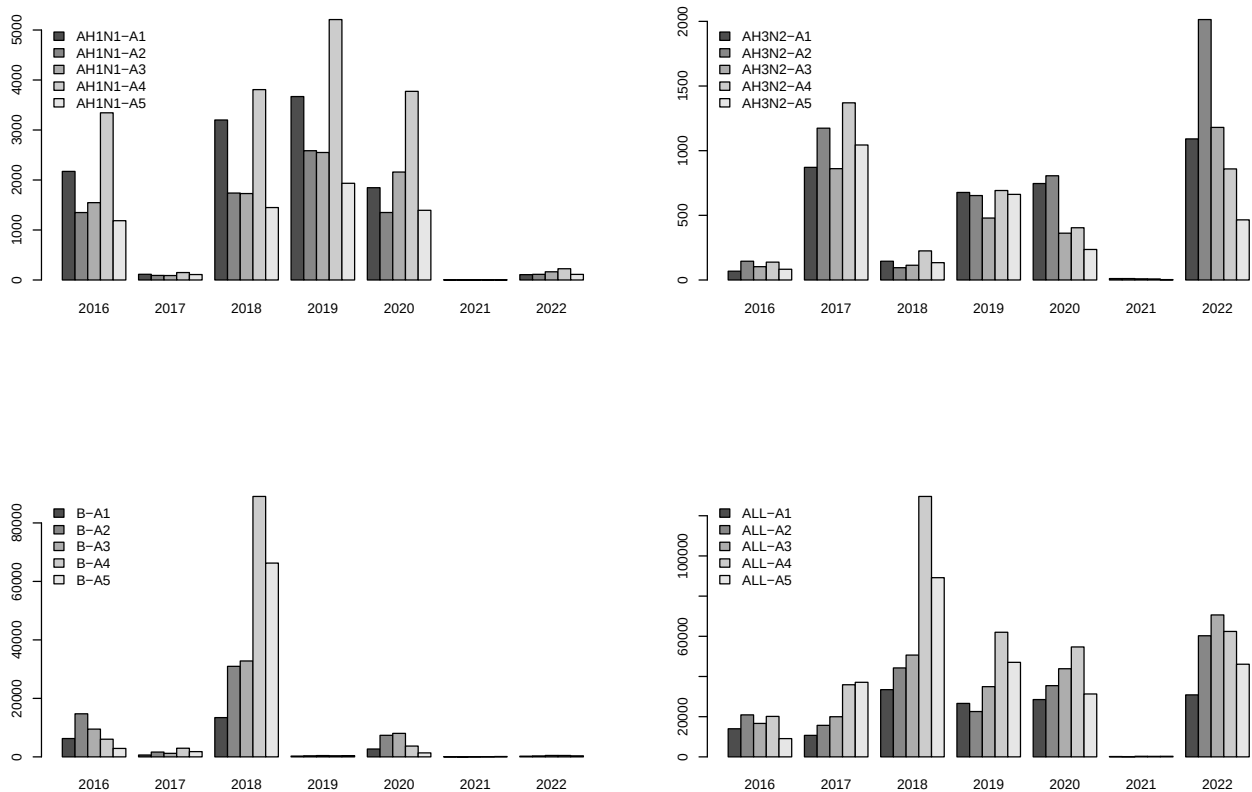


Fig.6 Number of annual cases of lineages AH1N1 (topleft), AH3N2 (topright), B (bottomleft) and Influenza (bottomright) by age groups A1-A5 during the period 2016-2022.

Variable	Type	Description
YEAR	integer	Reporting Year (ISO 8601)
A1	integer	Number of AH1N1 annual cases by age-groups A1-A5
A2	integer	Number of AH3N2 annual cases by age-groups A1-A5
A3	integer	Number of B (all subtypes) annual cases by age-groups A1-A5

A4	integer	Number of Influenza (all types) annual cases by age-groups A1-A5
A5	integer	Number of AH1N1 annual cases by age-groups A1-A5
Tab.5 Short description for the selected variables from the RKI-D3 dataset.		

Data ID	Source	Frequency	Period	Variables
W-D1.csv W-D2.csv	Weidemann et al	Weekly (43 weeks)	2004-2019	n=4
Weekly information defaults to week 1 of the new year of a season (negative weeks are the weeks of the old year of a season). The year refers to the new year of a season, i.e. 2004 is the 2003/04 season. The 2009/10 season is missing as a pandemic season and the age-groups are not the same between the two Sentinel data deliveries (5 age-groups), once with 6, sometimes with 7 age-groups. Additionally, It was not possible to integrate the data provided by Dr. Weidemann's Group with the other databases. Its sampling consists of 43 weeks in the year and not 52, like other databases. Finally, the period covered by the database, covers only the period 2004 to 2019, different from that will be required in the experiment, which will cover the period from 2016 to 2022.				

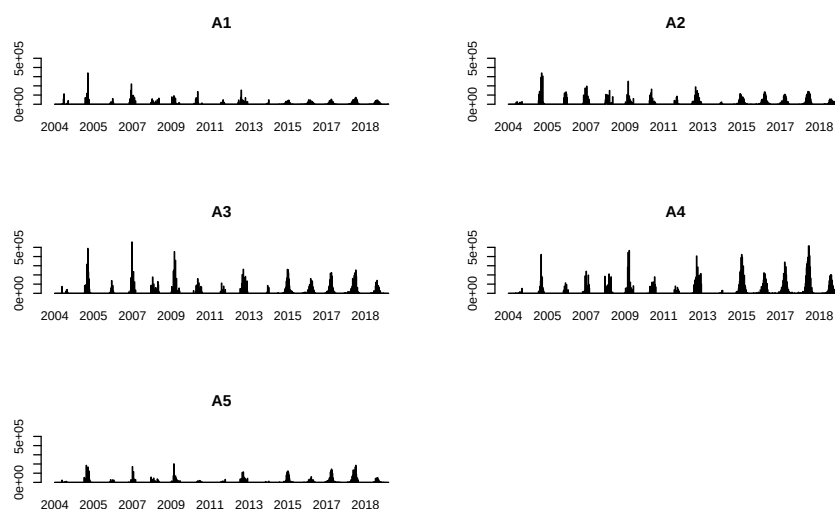
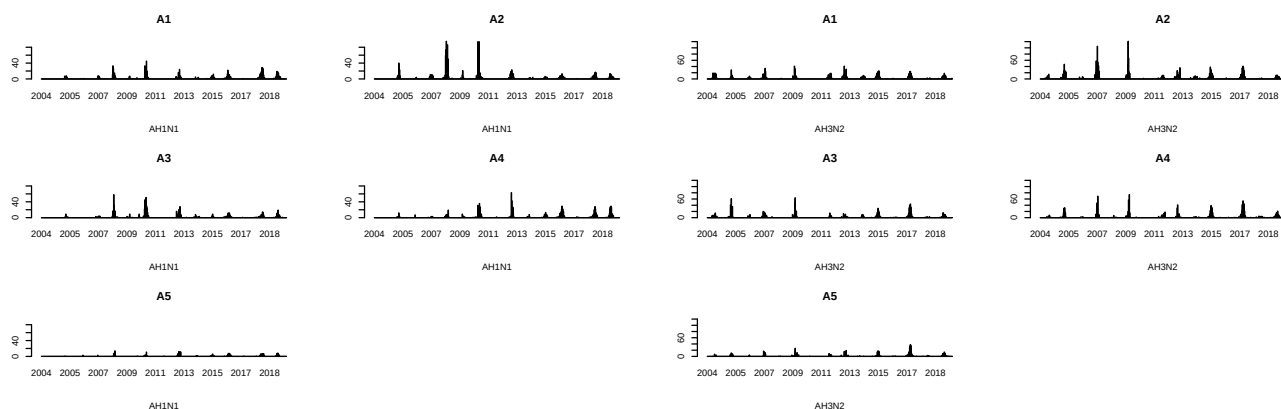


Fig.12 Number of consultations by age group (W-D1) during the period 2004-2019.

Variable	Type	Description
YEAR	integer	Year (ISO 8601)
WEEK	integer	Week (ISO 8601)
A1	integer	Number of consultations by age-group A1(0-4)
A2	integer	Number of consultations by age-group A2(5-14)
A3	integer	Number of consultations by age-group A3(15-34)
A4	integer	Number of consultations by age-group A4(35-59)
A5	integer	Number of consultations by age-group A5(60+)

Tab.5 Short description for the variables of the W-D1 dataset.



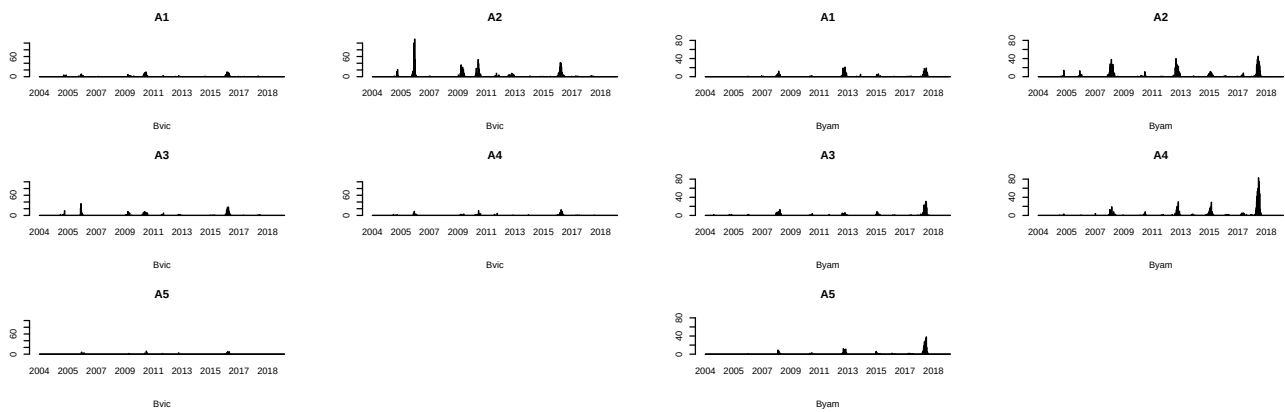


Fig.13 Number of weakly registered cases of lineages AH1N1 (topleft), AH3N2 (topright), Bv (bottomleft) and By (bottomright) by age group during the period 2004-2019.

Variable	Type	Description
YEAR	integer	Year (ISO 8601)
WEEK	integer	Week (ISO 8601)
AH1N1	integer	Number of AH1N1 lineage detections by age-group A1(0-4)
AH3N2	integer	Number of AH3N2 lineage detections by age-group A2(5-14)
Bv	integer	Number of Bv detections by age-group A3(15-34)
By	integer	Number of By detections by age-group A4(35-59)

Tab.6 Short description for the variables of the W-D2 dataset.

2.2 Demographic Data

Data ID	Source	Data Collection Frequency	Period	Variables
G-POP.csv	GENESIS	Annually	1970-2023	n = 1

GENESIS (Gemeinsames neues statistisches Informationssystem) is a metadata-based statistical information system which has been developed together by the Federal Statistical Office of Germany and the statistical offices of the German Länder. The Federal Statistical Office of Germany has developed GENESIS-Online to give web-based access to GENESIS, that is, to provide the opportunity to use the statistical information system via the Internet.

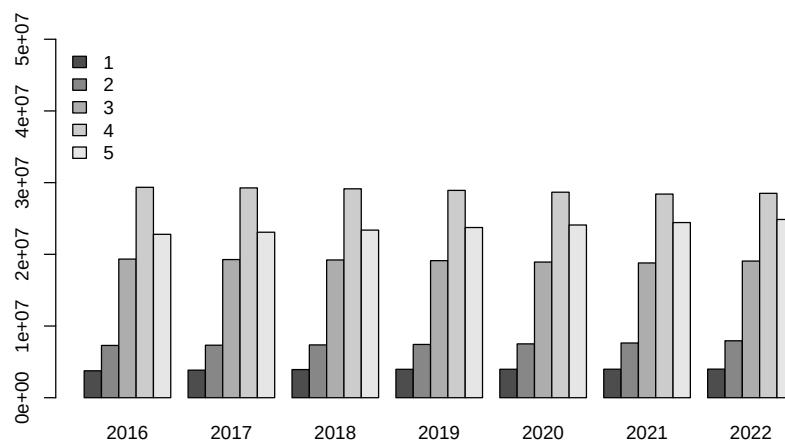


Fig.18 Number of people by age-groups A1-A5 for the period 2014-2022.

Variable	Type	Description
YEAR	integer	Year (ISO 8601)
GENESIS-POP-A1	integer	Number of people belonging to the A1 age-group
GENESIS-POP-A2	integer	Number of people belonging to the A2 age-group
GENESIS-POP-A3	integer	Number of people belonging to the A3 age-group
GENESIS-POP-A4	integer	Number of people belonging to the A4 age-group
GENESIS-POP-A5	integer	Number of people belonging to the A5 age-group

Tab.6 Short description for the variables of the G-POP dataset.

2.3 Wheater Data

Data ID	Source	Data Collection Frequency	Period	Variables
Wetter.csv	Visual Crossing	Weekly	2010-2023	1

Visual Crossing is a leading provider of weather data and enterprise analysis tools to data scientists, business analysts, professionals, and academics.

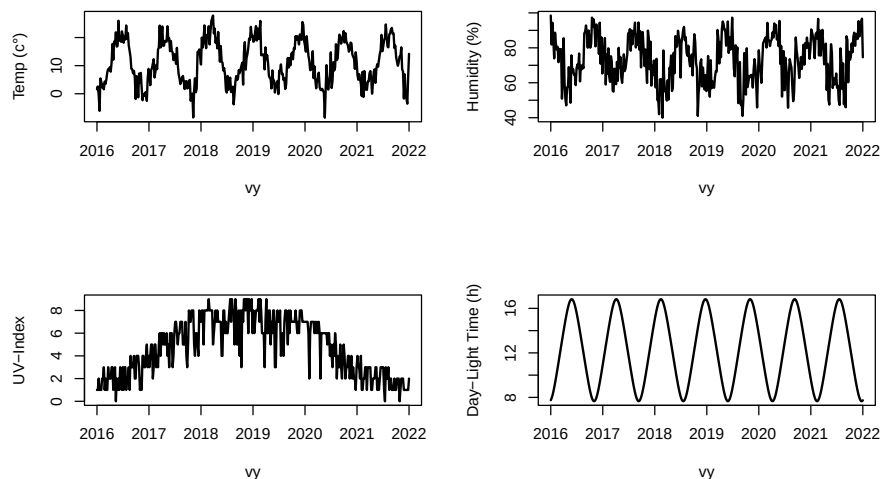


Fig.19 Some meteorological parameters (Temperature, Humidity, UV-Index and Daylight Time) from a set of 28 parameters (wetter) from the Visual Crossing database.

Variable	Type	Description	Variable	Type	Description
YEAR	integer	Year (ISO 8601)	winsspeed	numeric	Wind Speed (kph)
WEEK	integer	Week (ISO 8601)	winddir	numeric	Wind Direction (degrees)
tempmax	numeric	Maximum Temp (C°)	sealevelpressure	numeric	Sea Level Pressure (mb)
tempmin	numeric	Minimum Temp (C°)	cloudcover	numeric	Cloud Cover (%)
temp	numeric	Mean Temp (C°)	visibility	numeric	Visibility (km)
feelslikemax	numeric	Feels Like Minimum (C°)	solarradiation	numeric	Solar Radiation (W/m ²)
feelslikemin	numeric	Feels Like Maximum (C°)	solarenergy	numeric	Solar Energy(MJ/m ²)
feelslike	numeric	Feels like (C°)	uvindex	numeric	UV Index
humidity	numeric	Relative Humidity (%)	sunrise	time	Sunrise Time
precipcover	numeric	Precipitation Cover (%)	sunset	time	Sunset Time
preciptype	categorical	Rain, Snow, Freezing-rain	dew	numeric	Dew Point (C°)
snowdepth	numeric	Snow Depth (cm)	icon	categorical	A Weather Icon
windgust	numeric	Wind Gust (kph)	daylight-time	numeric	Duration of the Daylight

Tab.8 Short description for the variables of the Wetter dataset.

3 References

Fitzner J, Qasmieh S, Mounts AW, Alexander B, Besselaar T, Briand S, Brown C, Clark S, Dueger E, Gross D, Hauge S, Hirve S, Jorgensen P, Katz MA, Mafi A, Malik M, McCarron M, Meerhoff T, Mori Y, Mott J, Olivera MTDC, Ortiz JR, Palekar R, Rebelo-de-Andrade H, Soetens L, Yahaya AA, Zhang W, Vandemaele K. Revision of Clinical Case Definitions: Influenza-like Illness and Severe Acute Respiratory Infection. *Bull World Health Organ*. 2018 Feb 1;96(2):122-128. <https://dx.doi.org/10.2471/BLT.17.194514>

Mossong, J., Hens, N., Jit, M., Beutels, P., Auranen, K., Mikolajczyk, R., Massari, M., Salmaso, S., Tomba, G. S., Wallinga, J., Heijne, J., Sadkowska-Todys, M., Rosinska, M., & Edmunds, W. J. (2008). Social Contacts and Mixing Patterns Relevant to the Spread of Infectious Diseases. *PLoS medicine*, 5(3), e74. <https://doi.org/10.1371/journal.pmed.0050074>

Weidemann, F., Remschmidt, C., Buda, S. *et al.* Is the Impact of Childhood Influenza Vaccination less than Expected: A Transmission Modelling Study. *BMC Infect Dis* 17, 258 (2017). <https://doi.org/10.1186/s12879-017-2344-6>

World Health Organization (2017). Pandemic Influenza Severity Assessment (PISA): a WHO Guide to Assess the Severity of Influenza Epidemics and Pandemics. [Accessed September 7, 2023](#)